Download our 2025 World Air Quality Report now!

Measure Air Quality

PROTECT YOUR HEALTH & THE EARTH

Clean air is essential for life, as air pollution affects our health and our planet.

Measuring air quality is the first step toward protecting both human health and the environment.

Air pollution doesn't just make the air harder to breathe—it affects nearly every system in the body. According to the World Health Organization (WHO), it contributes to an estimated 7 million premature deaths worldwide each year, making it the greatest environmental threat to human health.

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EXPLORE POLLUTANTS LINKED TO HEALTH RISKS

Air pollution doesn't just affect people—it also disrupts Earth's climate systems and ecosystems.

Burning fossil fuels, such as coal, oil, and natural gas, releases greenhouse gases, including carbon dioxide (CO_2), methane (CH_4), nitrogen dioxide (NO2), and sulfur dioxide (SO_2). These gases trap heat in the atmosphere, driving climate change and contributing to ground-level ozone and particle pollution.

The same combustion processes also create particulate matter (PM), including black carbon and brown carbon, which absorb sunlight and accelerate global warming.

Together, these pollutants trap heat, intensify wildfires, and drive more extreme weather—creating feedback loops that harm both the environment and human health.

HOW AIR POLLUTION DRIVES CLIMATE CHANGE

The main types of air pollutants—and what they mean for your health

The Air Quality Index (AQI) simplifies complex air data into a single number that reflects health risk—from Good to Hazardous.

IQAir's AQI+ tracks six key pollutants:

Both PM10 and PM2.5 are inhalable particles, but their size determines how deeply they penetrate the body and how they affect health.

Particles smaller than 10 microns

Coarse dust, pollen, and mold spores. These mostly lodge in the nose, throat, and upper airways, causing irritation, coughing, and worsening of allergies and asthma. They can reach the lungs but are less likely to cross into the bloodstream than PM2.5.

Particles 2.5 microns or smaller

Fine particles from combustion (vehicles, power plants, residential heating/cooking, wildfires) and secondary formation in the atmosphere when gases like SO_2, NO_x, and NH_3 react. PM2.5 bypasses the body's defenses, reaches deep into the lungs, and can enter the bloodstream, triggering inflammation of blood vessels and affecting the heart, brain, and other organs. PM2.5 often stays airborne longer and travels farther than larger particles, so it's more likely to drive AQI levels and public-health advisories.

While not a particle, ozone is a major AQI pollutant. It forms when sunlight reacts with nitrogen oxides (NO_x) and volatile organic compounds (VOCs).

Ozone irritates airways, reduces lung function, and increases asthma symptoms—especially on hot, sunny days.

PM2.5 often dominates air-quality readings because it is the smallest, most persistent, and most widespread pollutant measured by the Air Quality Index (AQI).

Unlike heavier particles that settle quickly, PM2.5 can remain suspended for days or weeks, traveling across cities and continents. It comes from nearly all major sources—vehicles, industrial combustion, wildfires, agricultural burning, and domestic heating and cooking—making it a universal indicator of poor air quality.

Because PM2.5 is so fine, it infiltrates buildings, bypasses many ventilation systems, and reaches deep into the lungs and bloodstream, driving both short- and long-term health risks. It is frequently listed as the "main pollutant" in AQI reports because it is most likely to exceed safety limits and affect the largest number of people.

LEARN MORE ABOUT PM2.5

There's no completely risk-free level of air pollution.

The World Health Organization (WHO) sets a guideline of ≤ 5 µg/m^3 (annual average PM2.5) to minimize long-term harm. Use this benchmark to gauge your long-term risk—short-term spikes still matter, especially for children, older adults, and people with respiratory conditions.

READ ABOUT WHO'S AIR QUALITY GUIDELINES

- **Step 1:** Download the AirVisual App to track your exposure wherever you go
- **Step 2:** Compare indoor and outdoor data to see how your air stacks up against healthy air guidelines

- **Step 3:** Take action with ventilation or filtration when levels rise
- **Optional:** Install an AirVisual Monitor for hyperlocal, real-time PM2.5 and AQI readings

EXPLORE IQAIR MONITORS

Want the big picture of air pollution?

The AirVisual App delivers historical, real-time, and forecast air pollution data at a glance**:**

EXPLORE THE AIRVISUAL APP →



Historical, real-time, and forecast air pollution data

Receive detailed figures on key air pollutants in more than 5,000 cities in over 100 countries.



7-day air pollution and weather forecasts

Plan your outdoor activities to optimize comfort and health.



Air quality alerts

Get notified when outdoor air quality isn't up to your standards.



Real-time air quality and weather information

Live global coverage of key air pollution and weather parameters.



Air pollution news and educational resources

Stay up to date on air pollution current events, medical findings and breaking news.



Health recommendations

Get personalized recommendations on helping to protect yourself from air pollution.

Using the AirVisual App, set Home, Work, and Outdoor locations to build a personal dashboard that tracks hourly air quality trends — helping you see patterns, spot risks, and stay protected.

EXPLORE THE AIRVISUAL APP

Clean air begins with small, informed actions that reduce pollution sources, bring in fresh air when it's safe, and filter what you can't avoid.

Learn how to improve your air quality and take control of your environment so you can create healthier spaces where you live, learn, and work.

EXPLORE WAYS TO IMPROVE AIR QUALITY

Help your community see the air they breathe.

Host an air quality station and share hyper-local data to IQAir's global map so neighbors, schools, and responders can act quickly when air quality changes.

BECOME A CONTRIBUTOR

Community-generated air quality data helps reveal where exposure is highest and who is most affected. Join our community of individuals and global citizen scientists contributing data to our free, real-time air quality platform.

JOIN THE CONTRIBUTOR COMMUNITY →



Individuals & citizen scientists

share neighborhood conditions and protect vulnerable loved ones



Schools & universities

integrate monitoring into education and campus safety





Hospitals & Clinics

support patient and staff safety and provide community alerts

Cities & agencies

extend official networks, improve coverage, enable data-driven responses

NGOs & community groups

raise awareness and advocate for clean-air action

Spotlighting partners using data for impact:

EXPLORE OUR CONTRIBUTORS →



FPT University in Vietnam

Campus-wide monitoring was integrated into curriculum and to inform and protect students and staff.



MEED Foundation in Iraq

Innovative regional network expanded public access to air quality data.

Read full story →

Read full story →



Langston in Wisconsin

When Langston Verdin went looking for asthma triggers in his city, he chose air quality monitoring for the job.

Read full story →

[1] In independent laboratory testing, the HyperHEPA filter was >99.9% effective at filtering Staphylococcus aureus bacteria, and the HealthPro series air cleaner with HyperHEPA filter was >99.9% effective at filtering Staphylococcus albus bacteria, and ≥99.52% effective at filtering Escherichia coli bacteria.

[2] In independent laboratory testing, the HealthPro series air cleaner with HyperHEPA filter achieved 99.9% reduction of the Influenza A (H1N1) virus, and >99.9% reduction of Human coronavirus HCoV-229E (ATCC VR-740), and >99.9% reduction of Human enterovirus 71 (ATCC VR-1432).

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